

Article

The Effect of Preliminary Mixing Methods on the Properties of PA6 Composites with Molybdenum Disulphide

Dmitry Zavrazhin *, Tatiana Dyachkova , Artem Obukhov , Mikhail Krasnyanskiy, Elena Burakova, Denis Dedov and Anastasia Chuprikova

Laboratory of VR simulators, Rectorate, Institute of Technology, Tambov State Technical University, Tambov 392000, Russia

* Correspondence: zavrazhin-do@yandex.ru

Abstract

This study is motivated by the severe tribological regime of PA6 composites in VR platforms operating under dry or boundary lubrication, where alternating shear during foot rotation, localised contact pressures, and third-body abrasion concurrently challenge wear resistance and retention of strength. This paper presents the results of research into the properties of composites based on polyamide PA6 and molybdenum disulphide, obtained by combining the components through high-intensity mechanochemical activation in a planetary mill and classical mixing in a turbulence mixer. We demonstrate that varying the energy of the premixing stage (mechanochemical activation versus low-energy premixing) serves as an effective means of interfacial engineering in PA6/MoS₂ composites, enabling simultaneous enhancement of mechanical and tribological properties at low filler contents. Analysis of experimental composite samples using Fourier-transform infrared spectroscopy (FTIR) indicates the interaction between MoS₂ and oxygen-containing groups of polyamide while maintaining its overall chemical composition. According to the TG-DSC curves, modification of polyamide leads to an increase in the melting temperature by 2 °C, while mechanical activation ensures stronger interaction between the matrix and the filler. Compared to pure PA6, the tensile strength of composites increases by 10–20% for mechanoactivated materials and by 5–10% for materials obtained by conventional methods. The mechanical activation effect is observed even at minimal amounts (0.25 and 0.5%) of MoS₂ in composites. The toughness of all composites, regardless of the mixing method, increases by 5–7% compared to pure polyamide. All composites show a 10–20% reduction in the coefficient of friction on steel. Simultaneously, the water absorption of composites becomes 5–20% higher than that of the original material, which indicates a change in structure and an increase in porosity. The obtained composite materials are planned to be used for manufacturing platforms for the movement of virtual reality (VR) operators.

Keywords: polyamide 6; molybdenum disulphide (MoS₂); mechanical activation; tribological properties; thermal stability; diffusion characteristics; mixing method



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1. Introduction

Virtual reality (VR) technologies are being used increasingly in the entertainment industry, as well as for training specialists in various industries. One of the issues is creating a platform that combines virtual reality (programme code) and an integrated omnidirectional track, ensuring the operator's safe movement under conditions close to real-world physics. For the production of such walkways, steel is used, which can be coated

with a material that has a low friction coefficient. At the same time, it is usually necessary to use special footwear with inserts made of easily sliding materials. Various polymers (ultra-high molecular weight polyethylene, fluoroplastic, and polyamide), film coatings, and substances that facilitate sliding can also be used.

To obtain products with high anti-friction properties, polymer materials reinforced with short fibres or various fillers are increasingly being used. However, they are characterised by relatively low toughness and resistance to external forces; their uniform surface is easily damaged by loads, resulting in dents and scratches. Such distortions to the shape and dimensions of the sliding elements can reduce the quality of interaction with the VR environment and cause injury to the operator.

One of the possible solutions to this problem is the development of new specialised polymer composite materials or the adaptation of existing ones.

Polyamide 6 (PA6) is a semicrystalline thermoplastic that is mechanically easy to process and has physical and mechanical characteristics that are ideal for use in mechanical engineering for various engineering tasks, in the creation of films, fibres and moulded products, and as a friction material [1,2].

Polyamide PA6 is also used as a matrix for various composite materials. Introducing nanoparticles of various types into PA6 significantly improves its tribological properties [3–5].

The mechanical properties of thermoplastics are crucial for assessing the vulnerability of moulded products to scratches and reflect the material's operational characteristics in terms of durability and friction behaviour [6]. For example, in [7] it is noted that surface shear strength and indenter contact area can be used to determine the friction force in a ball-on-disc test.

The coefficient of friction is the most important property of a surface. Adhesion theory postulates the coefficient of friction as the ratio of indentation hardness to shear strength [8]. Factors affecting friction and the endurance of PA6 products were identified by R. T. Steinbuch in [9].

Various fillers are added to polymers to improve their mechanical properties. Carbon materials (MWCNT, CF, graphite, etc.) are often used for this purpose [10]. Study [11] shows that the use of dispersed fillers based on potassium tetratitanate contributes to an increase in the degree of crystallinity of polyamide-6, which has a beneficial effect on its physical and mechanical characteristics. Studies [12–14] have shown that the properties of polyamides can be improved using micron-sized glass beads. Modification can also be carried out using various fibres. In work [15], optimisation of the tribological properties of glass fibre reinforced PA6 (GF/PA6, 15/85 by weight) was carried out for high-performance friction materials using single or combined solid lubricants, such as polytetrafluoroethylene (PTFE) and ultra-high molecular weight polyethylene (UHMWPE). Similar results were obtained in [16,17]: reinforcing the polymer matrix with glass fibre or carbon fibre effectively reduces the friction coefficient and wear rate.

Layered solid lubricants, such as MoS₂, are also effective additives in lubricants and polymers [9]. In [18], it is shown that the introduction of MoS₂ into the PA12 matrix improves the impact properties of the polymer. The friction coefficient and wear rate of the PA12/MoS₂ composite obtained by laser sintering were significantly reduced, and the surface of the material remained smooth during wear testing.

Some studies indicate the advisability of using mixed fillers containing molybdenum disulphide in composites. Thus, the authors [19] investigated a dry self-lubricating material based on these carbon and glass fibres with additives of polytetrafluoroethylene, graphite, and molybdenum disulphide. It has been established that the wear of filled polymer materials decreases, and its mechanism consists of the sequential destruction of the polymer

matrix first, followed by the reinforcing components, which leads to phase separation and destruction of the material in general.

In [20], it was found that short carbon fibre (SCF) and graphite flakes fillers can effectively increase the wear resistance and load-bearing capacity of base polymers. The results of the studies showed that the addition of carbon fibre was effective in reducing friction and wear in pure PA6, but the MoS₂ filler increased its wear. The reduction in wear and friction was more significant when carbon fibre was used as reinforcement together with the MoS₂ filler [21].

MoS₂ can be used as a dry lubricant, as a surface friction material in films, coatings, and composite materials. For example, in [22,23], MoS₂ coatings are formed using high-speed powder spraying. The addition of MoS₂ to a polymer matrix reduces the friction coefficient by approximately 70% [24–27].

The mechanical properties of polymer composites are determined not only by the filler's type and composition but also by its particle size. As shown in [28], nanoscale molybdenum disulphide (MoS₂) is the most effective.

The method of combining the components should have a significant effect on the particle size and distribution of the filler in the composite volume. However, information on the effect of the methods of introducing molybdenum disulphide into polyamide PA on the properties of the resulting material is very limited. Two methods are usually used to mix powder materials: (1) in mixers for dry materials; (2) in melt by adding powder modifiers to the polymer melting zone.

Classic mixing of MoS₂ powder with polyamide granules in batch-type devices ensures high productivity. Existing planetary mixers allow for mechanical activation of the modifier that improves the final product's properties [29,30]. High-energy processing methods in a planetary mill enable mechanical synthesis of composite materials as a result of intensive cold-welding of particles [31].

When obtaining PA6-based rolled material blanks, the technology used to mix powder materials is of interest, since free casting is a widely used method of obtaining blanks. In this case, it is important to ensure the modifier is distributed evenly throughout the polymer.

The analysis clearly shows that, to obtain composites with improved characteristics, more effective methods of mixing polymer composite components are necessary. This study aimed to investigate how the mixing method affects the properties of polyamide 6 and molybdenum disulphide composites.

Beyond lamellar solid lubricants, interface manipulation via two-dimensional fillers (e.g., MXenes) has been reported for PA6 to tailor surface porosity and mass transport. However, such routes typically require chemical functionalisation and complex dispersion protocols and often target permeation rather than dry-friction durability. Given our focus on scalable, low-complexity processing for tribologically loaded PA6 parts, MXene-based approaches are outside the present scope [32].

The study implements a process-centred approach to interphase control: under otherwise identical conditions, only the premixing stage is varied, while the extrusion finish is kept constant. This enables a direct comparison of how energy input affects the interphase structure and the resulting balance of properties.

We deliberately selected VR platforms as the main testing ground, since they impose on polymer composites one of the most demanding yet widely representative combinations of tribological and mechanical requirements: dry or boundary friction without lubrication, alternating shear loads during foot rotation, localised contact pressures, and inevitable abrasive contamination (dust, quartz particles, and rubber debris from footwear) [33,34].

The accompanying photographs (Appendix A, Figure A1) clearly show the characteristic damage patterns—a dense radial field of scratches in the central zone and large-scale

groove marks—that develop after multiple cycles of foot placement and turning. This loading mode serves as an explicit test of both wear resistance and the material's ability to retain strength and hardness at a low coefficient of friction.

From a practical perspective, the VR platform functions as a natural test rig with highly repeatable conditions and rapid accumulation of operational cycles, allowing the durability of materials to be assessed efficiently. At the same time, the chosen case study does not limit the broader relevance of the results: the identified causal relationships and the optimum low-dosage range of the solid-lubricant filler (MoS_2) are directly transferable to any dry-sliding PA6 components—bushings, guides, bearing liners, and sliding surfaces used in rehabilitation and transport systems as well as in mechanical engineering, where the retention of strength under high friction is of critical importance [35].

Thus, the focus on VR applications is justified by their representativeness and the severity of their operating conditions, which provide a rigorous and reproducible framework for verifying materials-science conclusions.

2. Materials and Methods

Polyamide PA6 grade «Volgamid 27» (PJSC Kuibyshev Azot, Tolyatti, Russia), corresponding in its characteristics to polyamide-6 grade 210/310, was used to manufacture the experimental samples. Unfilled general-purpose PA6 was used; according to the manufacturer's datasheet, it is a base polymer for compounding and BCF filament production. The key certified parameters are as follows: relative viscosity (96% H_2SO_4 , 1 g/100 mL) 2.75 ± 0.05 ; mass fraction of extractable substances $\leq 0.6\%$; moisture content $\leq 0.06\%$; melting temperature $\geq 215^\circ\text{C}$. The melt flow rate (MFR) for this grade is not specified by the manufacturer; therefore, the work relied on the datasheet parameters and controlled material moisture during processing.

Molybdenum disulphide DMI-7 (LLC GK CMM, Moscow, Russia), compliant with technical specification TU 48-19-133-90, was used as the filler; for this grade the normative specifies that particles $\leq 7\ \mu\text{m}$ account for $\geq 96\%$, hence $D_{90} \leq 7\ \mu\text{m}$ (the oversize fraction is $\leq \sim 4\%$).

The components were mixed in two ways.

(1) In a laboratory planetary ball mill Activator 2SL (LLC Chemical Engineering Plant, Novosibirsk, Russia), molybdenum disulphide was pretreated for 2 min at a planetary disc rotation speed of 1000 rpm and a drum rotation speed of 1500 rpm. Then polyamide PA6 granules were added to the drums, and the components were mixed under the same conditions. The volume of one drum is 250 mL. The mixing drums were equipped with a water-flow cooling system, maintaining the barrel temperature close to ambient; by the end of processing, the batch temperature did not exceed the surrounding temperature by more than approximately $5\text{--}10^\circ\text{C}$. No signs of softening or pellet agglomeration were observed. The samples obtained by this method are labelled as MA.

(2) The components were mixed in a turbulent mixer for bulk components of periodic action C 2.0 of the «drunken barrel» type Drum Tumblers (LLC Vibrotehnik, St. Petersburg, Russia). The useful volume of the bowl was 1.7 cubic decimetres, the bowl rotation speed was 50 rpm, and the mixing time was 10 min. The samples obtained by this method are labelled as DT in the paper.

The MoS_2 content in mixtures with polyamide PA6 varied in the range from 0.25 to 5 mass percentage.

The final test specimens were produced by injection moulding on a «Vector» vertical injection machine (Russia) at a material cylinder temperature of 225°C , an injection rate of 50 mm/s, and a holding pressure time of 5 s. Specimen moulding was performed in

accordance with ISO 294-1:2017 [36], and sample preparation and conditioning followed ISO 291:2008 [37] (standard atmosphere of 23 ± 2 °C and 50% relative humidity).

Table 1 presents the list of experimental samples obtained and their designations.

Table 1. List and designation of experimental samples.

Sample Designation	Composition	Mixing Mode
0.25 MA	PA6 + 0.25% MoS ₂	Mechanical activation in planetary mills
0.5 MA	PA6 + 0.5% MoS ₂	
1 MA	PA6 + 1% MoS ₂	
3 MA	PA6 + 3% MoS ₂	
5 MA	PA6 + 5% MoS ₂	
0.25 DT	PA6 + 0.25% MoS ₂	Turbulent mixer drum tumblers
0.5 DT	PA6 + 0.5% MoS ₂	
1 DT	PA6 + 1% MoS ₂	
3 DT	PA6 + 3% MoS ₂	
5 DT	PA6 + 5% MoS ₂	

Photographs of the modified PA6 sample surfaces after tribological testing were obtained using the METAM RV-34 optical microscope (Russia).

The physical and mechanical characteristics were measured on a universal testing machine UTS 101-5 (LLC Test-systems, Ivanovo, Russia). Tensile strength and elastic modulus under uniaxial tension were determined in accordance with ISO 527-1:2019/ISO 527-2:2012 [38,39]. Shear strength was evaluated using the procedure described in the Methods section (no universal ISO exists for solid thermoplastics; the loading scheme is specified herein).

Hardness was determined with an IT 5069 durometer, Shore D scale (LLC Impuls, Ivanovo, Russia), in accordance with ISO 868:2003 [40].

IR spectra were recorded on a Jasco FT/IR 6700 (Jasco International Co., Ltd., Tokyo, Japan) in ATR mode on a diamond crystal [41].

Thermogravimetric analysis (TGA) was performed using a NETZSCH STA 449 F3 Jupiter (NETZSCH-Feinmahltechnik GmbH, Selb, Germany) in air from 40 to 900 °C at a heating rate of 10 °C·min^{−1}, in accordance with ISO 11358-1:2022 [42].

Tribological characteristics were evaluated using a cylinder–steel disc configuration on an MI-2/2101TP tribometer (LLC Metroteks, Moscow, Russia) configured for hard plastics, following ISO 7148-2:2012 [43].

Water absorption of raw materials and composites was measured in accordance with ISO 62:2008 [44]. Water absorption was determined after moisture saturation of samples in boiling water for 30 min.

3. Results and Discussion

3.1. Visual Assessment of PA 6 Mixtures with MoS₂

The results illustrate process-enabled interface engineering: increasing the energy input at the premixing stage (mechanochemical activation, MA) promotes the formation of a more effective interphase at low MoS₂ loadings, leading to a simultaneous improvement in strength and stiffness alongside a reduction in the coefficient of friction.

When the components are mixed in different modes, the polyamide granules are dusted with molybdenum disulphide powder. It should be noted that at a certain mixture composition, MoS₂ begins to visibly fall off the polymer granules (Figure 1). In DT and MA samples, this begins to occur at a molybdenum disulphide content of 3 and 5 mass percentages. Filler shedding can lead to an uneven distribution of its particles within

the composite volume, which usually has a negative impact on physical and mechanical characteristics.

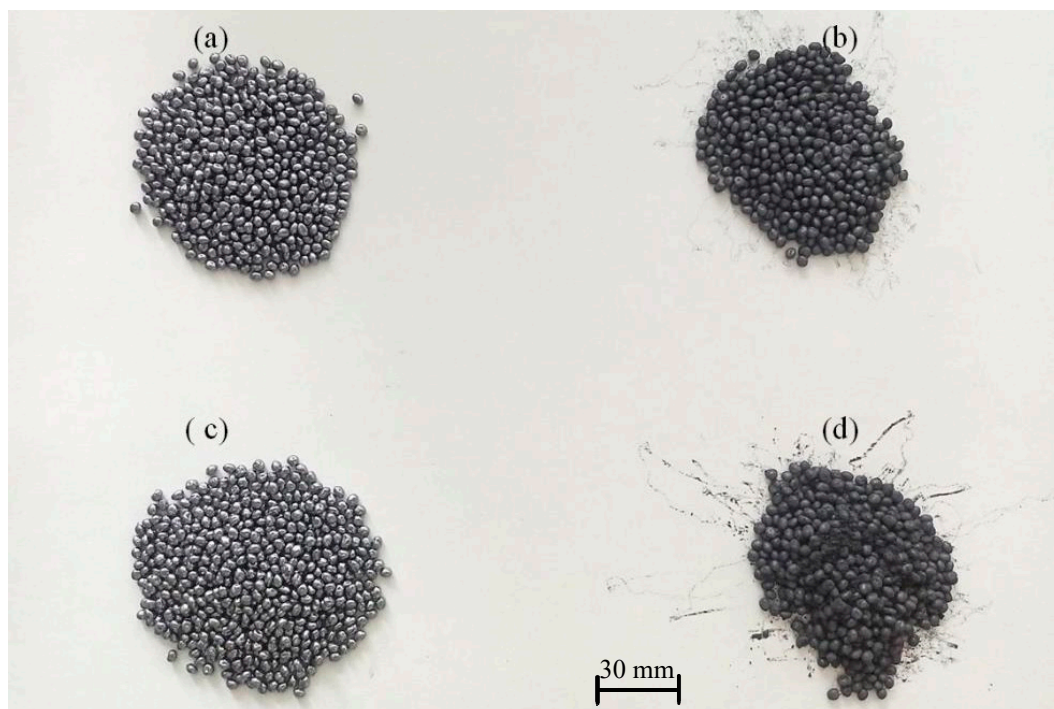


Figure 1. General view of the mixture samples: (a) 0.5 MA, (b) 0.5 DT, (c) 5 MA, (d) 5 DT.

Thus, mechanical activation allows a larger amount of filler to be applied to the surface of the polymer granules. At the same time, the granule sizes of the DT and MA samples are approximately the same, which means that this fact is not related to an increase in the specific surface area of the polymer but because of the possibility of mechanochemical reactions and mechanical alloying of components under high-intensity loading in a planetary mill.

The layered/multichannel, co-extrusion, and masterbatch routes can provide additional control over dispersion and interfacial adhesion but are associated with greater process complexity. In this study, we isolated the pure effect of premixing as a simple and scalable lever of interfacial engineering. The resulting “process → interphase → properties” map serves as a starting point for the integration of more advanced architectures—including multilayer channel structures and side-feeding techniques—in future research.

3.2. Examination of Composite Samples

The FTIR spectrum of the PA6 polyamide sample (Figure 2) appears typical. It exhibits peaks caused by vibrations of N-H, C-H, C=O, and C-N bonds. The presence of C-O bonds may be attributed to oxidation processes on the material surface. When molybdenum disulphide is introduced into the polymer, no significant changes in the IR spectrum occur. Only a slight decrease in the intensities of spectral lines can be noted as the concentration of the modifier increases, and the peaks characteristic of MoS₂ are not detected, probably due to their low intensity relative to the polymer. In the spectra of composite samples, apart from 0.5 MA, there is no peak at 1740 cm^{−1}, which may indicate the shielding of some carbonyl groups by molybdenum disulphide. In addition, interaction with other oxygen-containing groups is also probably occurring, as indicated by changes in the intensity and position of peaks caused by C-O bond vibrations. Furthermore, the interaction between molybdenum disulphide and polyamide is indicated by a change in the position of peaks

caused by vibrations of amide bonds (C-N). However, based on IR spectra, it cannot be said that there are any significant changes in the composition of the polymer material when MoS₂ is introduced.

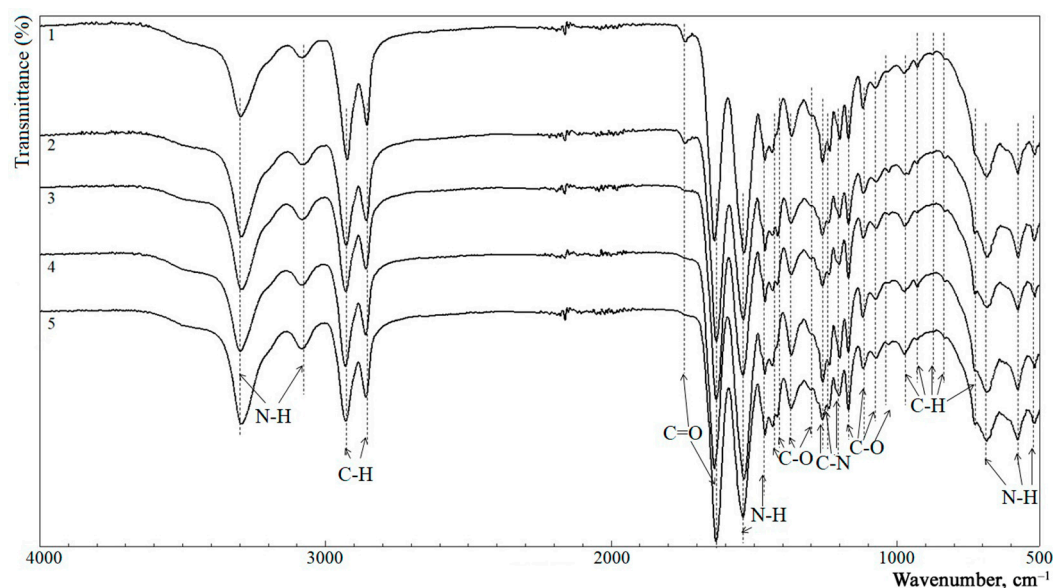


Figure 2. FTIR-spectra of samples PA 6 (1), 0.5 MA (2), 0.5 DT (3), 3 MA (4), and 3 DT (5).

The TG curves of the polyamide, the composites, and the PA6/MoS₂ variants (Figure 3) are similar in pattern. The main stage of mass loss, approximately about 90%, occurs within the temperature range of 350–500 °C, corresponding to the destruction of the PA6 polymer chains. With an increase in the concentration of molybdenum disulphide as a non-combustible component, there is a slight increase in the residual mass of the materials (Table 2).

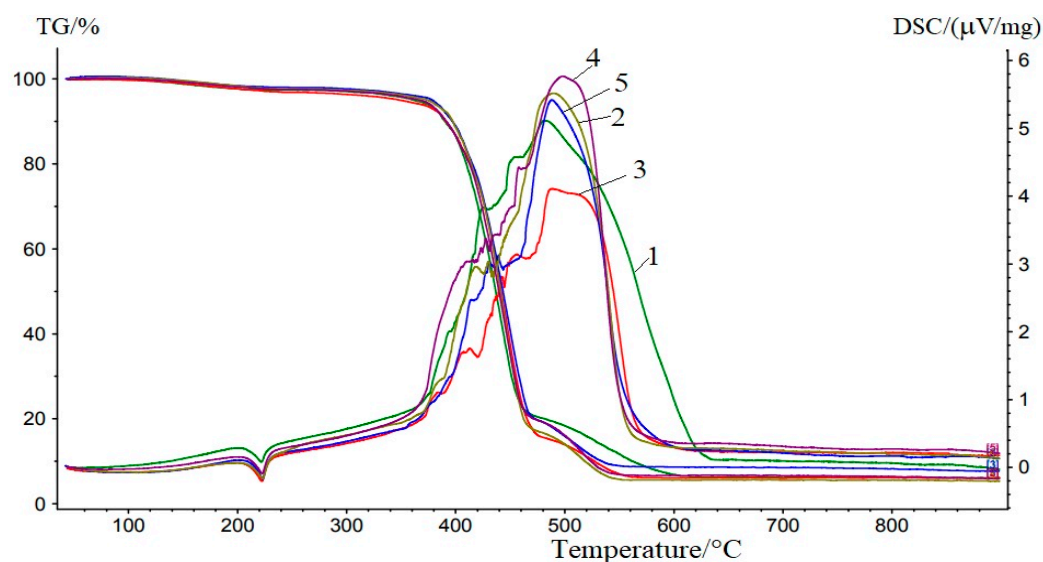


Figure 3. TG/DSC curves of samples PA6 (1), 0.5 MA (2), 0.5 DT (3), 3 MA (4), and 3 DT (5).

A 5% mass loss of neat PA6 is observed at 362 °C. The introduction of 0.5 wt% molybdenum disulphide (MoS₂) under the DT regime lowers this temperature to 349 °C, indicating a weakening of the intermolecular bonds within the polymer and limited interaction with the filler. However, when the MoS₂ concentration is increased to 3 wt% under the same mixing regime, the temperature corresponding to a 5% mass loss rises to 368 °C.

Table 2. Results of thermogravimetric analysis of the samples.

Sample	Temperature, °C		Residual Weight, %
	5% Weight Loss	50% Weight Loss	
PA6	362	437	5.7
0.5 MA	375	442	6.2
0.5 DT	349	441	6.1
3 MA	379	444	6.9
3 DT	368	440	7.7

Under mechanochemical activation (MA), the thermal stability of the composites improves further: 5% mass losses are observed at 375 °C and 379 °C for 0.5 wt% and 3 wt% filler contents, respectively.

The temperatures corresponding to a 50% mass loss show only a slight increase with the introduction of the filler in all cases; however, the MA specimens exhibit somewhat greater stability than those produced under the DT regime.

Thus, it can be inferred that during composite formation, both the incorporation of molybdenum disulphide and mechanochemical activation promote crosslinking between polyamide chains. In contrast, when mixing is carried out in the DT mode, achieving a comparable effect requires a higher filler content.

The DSC curves show a minimum corresponding to the phase transition (melting) of materials and a broad peak with several local maxima, which illustrates the stage of oxidative thermal destruction of the analysed samples. The presence of MoS₂ increases the temperature of the onset and most intense stage of decomposition. As the filler content increases from 0.5 to 3%, the corresponding temperatures increase by 5–10 °C.

The melting point (T_m) of pure PA6 is observed in the range of 220–225 °C. The addition of MoS₂ reduces the intensity of the endothermic melting peak and contributes to an increase in T_m by approximately 2 °C, regardless of the method of combining the components.

The section of the DSC curve of pure PA6 corresponding to thermal degradation is found in the temperature range from 370 to 640 °C. There are several local maxima corresponding to the degradation of polymer chains with different molar masses. The highest peak (T_{max}) is observed at a temperature of 490 °C. The introduction of molybdenum disulphide contributes to an increase in T_{max} by ≈ 10 °C, which may indicate the cross-linking of polyamide macromolecules. In addition, the nature of polymer degradation changes slightly, which usually indicates differences in the structure of the composite and the source material. Furthermore, an increase in MoS₂ concentration affects differently on the DSC curves of MA and DT samples. In the case of mechanically activated samples, a 0.5% modifier content reduces the thermal effect of material destruction, while a 3% modifier content increases it. At temperatures above 500 °C, oxidation of molybdenum disulphide is possible, during which additional heat is released. It can be assumed that an increase in the thermal effect of material destruction will be observed if molybdenum disulphide is not bound to polymer macromolecules. Therefore, in mechanically activated samples with a 0.5% content of MoS₂, strong interaction between the filler and the matrix is observed. The DT sample preparation mode does not provide such interaction even at low concentrations of molybdenum disulphide, so the thermal effect of their thermo-oxidative degradation is always higher than that of the initial polymer.

The results of experiments on uniaxial tensile testing of samples of the original polyamide and composites based on it are presented in Figure 4. Analysis of the tensile curves allows us to conclude that all materials demonstrate yield and subsequent cold drawing, which is common for semicrystalline plastics.

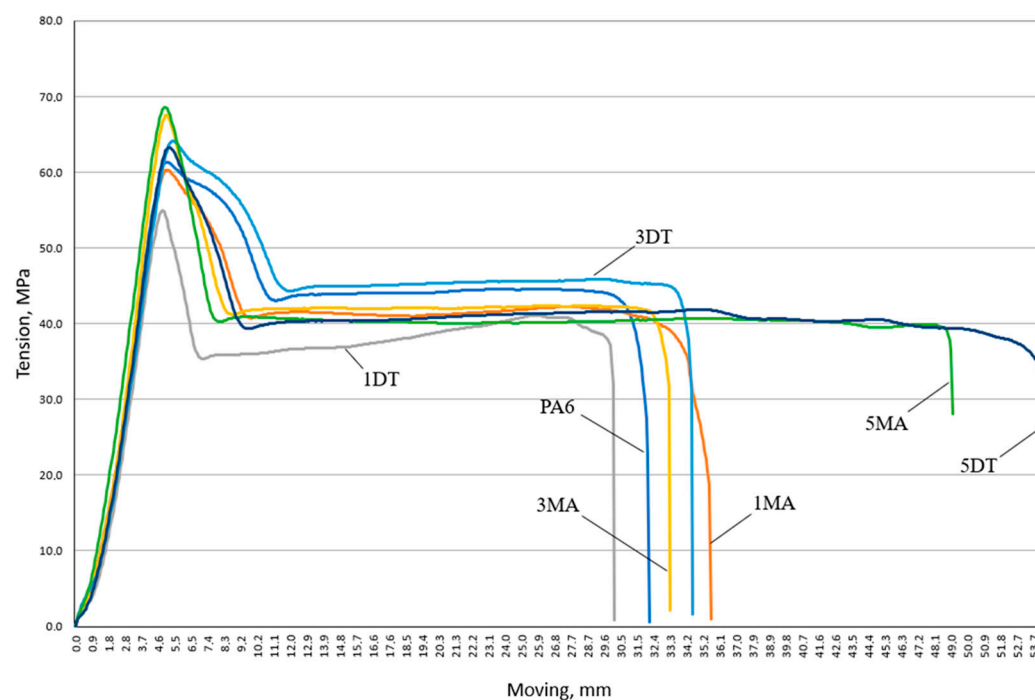


Figure 4. Diagrams of tensile strength and relative extension of PA6 samples and composites based on it under uniaxial tension.

The elastic modulus, tensile strength and relative extension at break are shown in Table 3. The addition of a modifier to the polymer matrix significantly changes the strength characteristics of the materials. The strength of samples obtained using mechanical activation increases by from 10 to 20%. The strength of composites obtained by conventional mixing of components is slightly lower, demonstrating a 5–10% increase compared to the original PA6. It should be noted that the effect of adding the MoS₂ modifier in the mechanically activated mode, even in small volumes (0.25 and 0.5%), achieves a 15% increase in composite strength. Samples obtained by the classical mixing method show a slight decrease in strength for 0.25% MoS₂ and an increase of up to 10% for 0.5% MoS₂.

Table 3. Results of physical and mechanical tests under uniaxial tension.

MoS ₂ Concentration, Mass Percentage	Tensile Strength, MPa		Relative Extension at Tear, %		Tensile Elasticity Modulus, MPa	
	MA	DT	MA	DT	MA	DT
0	64		72		1562	
0.25	70.5	63.5	75	77	1544	1458
0.5	70.8	68.6	74	76	1589	1508
1	67.5	56.6	74	71	1637	1602
3	76.2	68	57	55	1528	1524
5	73	70.9	61	62	1768	1722

The measurement uncertainties were as follows: for the tensometer, $\pm 1\%$; for the tensile modulus, $\pm 5\%$; for the tensile strength, $\pm 5\%$; and for the elongation at break, up to $\pm 10\%$. The mean value of each parameter was determined from the results of ten repeated tests.

A slight decrease in the strength of materials obtained in MA and DT modes when adding 1% MoS₂ can be explained by the occurrence of structural defects as a result of insufficient interaction between the modifier and the polymer matrix. The increase in strength characteristics for sample 3 MA is probably due to a fairly strong interaction between the components of the system.

The relative extension of composite samples from both systems with MoS₂ concentrations ranging from 0.25 to 1% increases slightly.

Samples obtained by the classical method of mixing powder materials using Drum Tumblers behave unevenly; the effect is achieved only at MoS₂ concentrations of 0.5, 3, and 5%. Simultaneously with the increase in strength, the relative extension decreases.

The recorded non-monotonic trends in strength, elongation, and modulus are interpreted as the result of competition between two opposing contributions. At low MoS₂ contents, the mechanochemically activated system exhibits improved interfacial load transfer (FTIR: weak but systematic changes in amide and carbonyl bands; DSC: shifts in T_m , T_{max} , and residual crystallinity), leading to gains in both strength and stiffness. With further increases in filler concentration, the proportion of aggregates and defects rises, causing local stress concentrations and a reduction in the reinforcement efficiency, which accounts for the observed “wavy” behaviour of the curves.

It may be inferred that strong interaction between the components as a result of their mechanical activation leads to significant changes in the physical and mechanical properties of composites when a minimal amount of modifier is added. As a result of mechanical activation, new bonds appear and a different composite structure is formed.

Similar conclusions can be reached when studying the Shore hardness of the composites (Table 4). Even at a minimum MoS₂ content (0.25%), the hardness of the material increases by 5%, but this indicator does not change with a further increase in the concentration of the modifier. At the same time, samples obtained with preliminary mechanical activation have slightly higher hardness values compared to samples obtained by the classical method.

Table 4. The influence of modifier concentration and component mixing method on the Shore hardness of the composites.

MoS ₂ Concentration, Mass Percentage	Hardness Index	
	MA	DT
0		74
0.25	78	77
0.5	77	76
1	78	78
3	78	77
5	78	78

MoS₂ particles (zone 1 in Figure 5) form a framework that creates a protective layer on the polymer surface.

The results of tribological tests of experimental samples are shown in Table 5. The addition of MoS₂ consistently reduces the friction coefficient of composites. However, for DT materials, the reduction is about 20%, while MA materials show an uneven reduction in the friction coefficient of 5–15%. This can be explained by the fact that at low MoS₂ loadings, a continuous transfer film is formed, exhibiting a reduced rate of degradation, whereas at higher concentrations, the likelihood of aggregation and micro-ploughing increases. These processes disrupt the integrity of the film and lead to a deterioration of the tribological performance. This interpretation is consistent with the increase in water absorption (an indirect indicator of matrix disorder or microporosity) shown in Figure 6, as well as with the non-monotonic behaviour of the mechanical properties presented in Table 3.

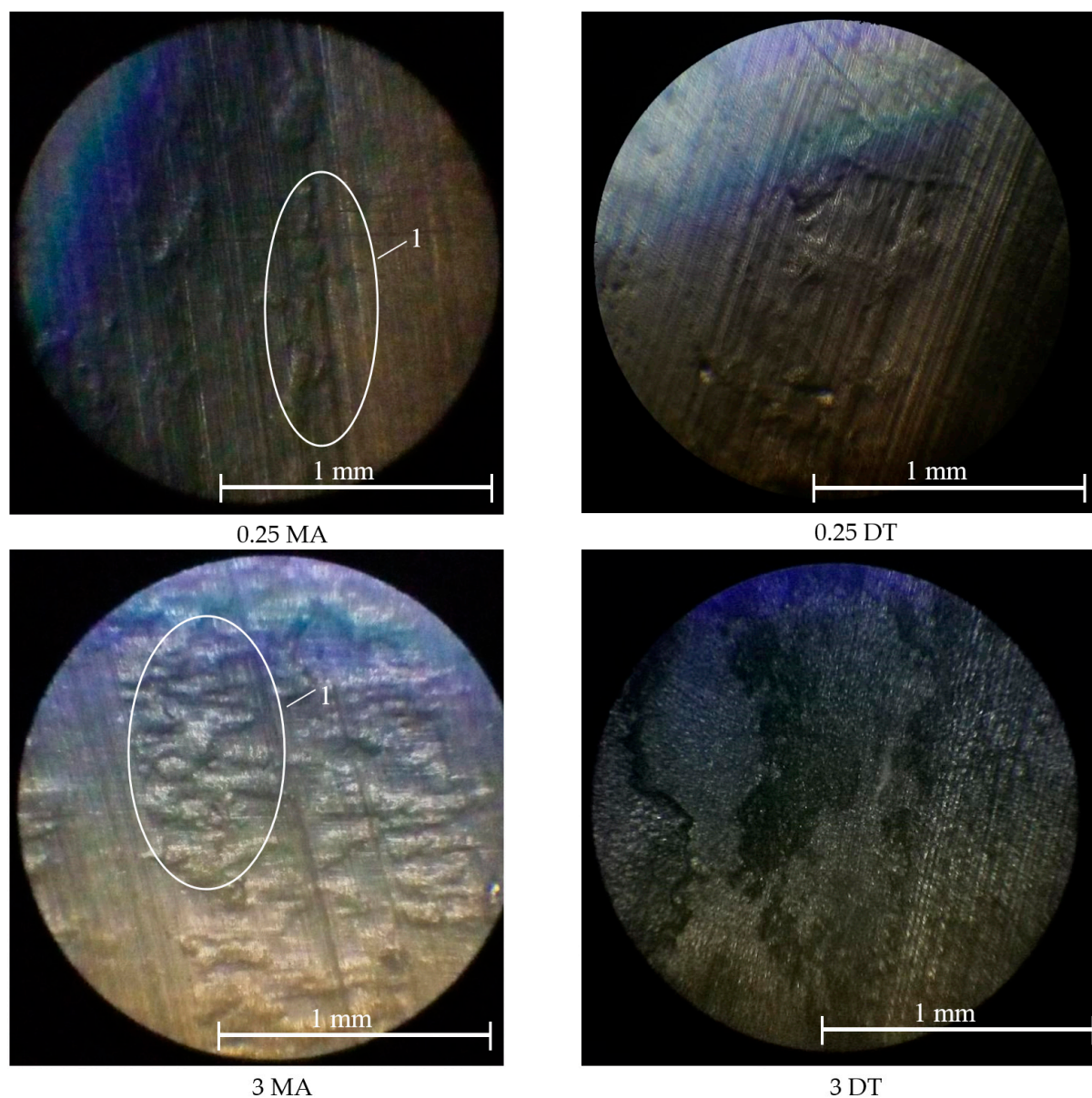


Figure 5. Micrographs of the composite sample surfaces after tribological testing.

Table 5. Results of tribological tests (coefficient of friction on steel).

The Content of MoS ₂ in the Composite, %	MA	DT
0	0.196	
0.25	0.168	0.151
0.5	0.161	0.157
1	0.174	0.157
3	0.174	0.169
5	0.186	0.156

The application of powder modifiers leads to a reduction in the density of the polymer matrix structure [45–47]. As a result, the porosity of the materials increases, which can be assessed by the amount of water absorption. Research into the water absorption of polymer composites has shown that even the introduction of insignificant amounts of MoS₂ additives significantly affects maximum water absorption. The obtained data (Figure 6) indicate a slight increase in the maximum water absorption of the polymer composite

with an increase in the concentration of the modifying additive, which is explained by the loosening of the polymer matrix due to the introduction of the powder modifier and the appearance of a certain number of defects in the structure of the sample. The increase in water content in composites reaches up to 20% compared to the initial PA6.

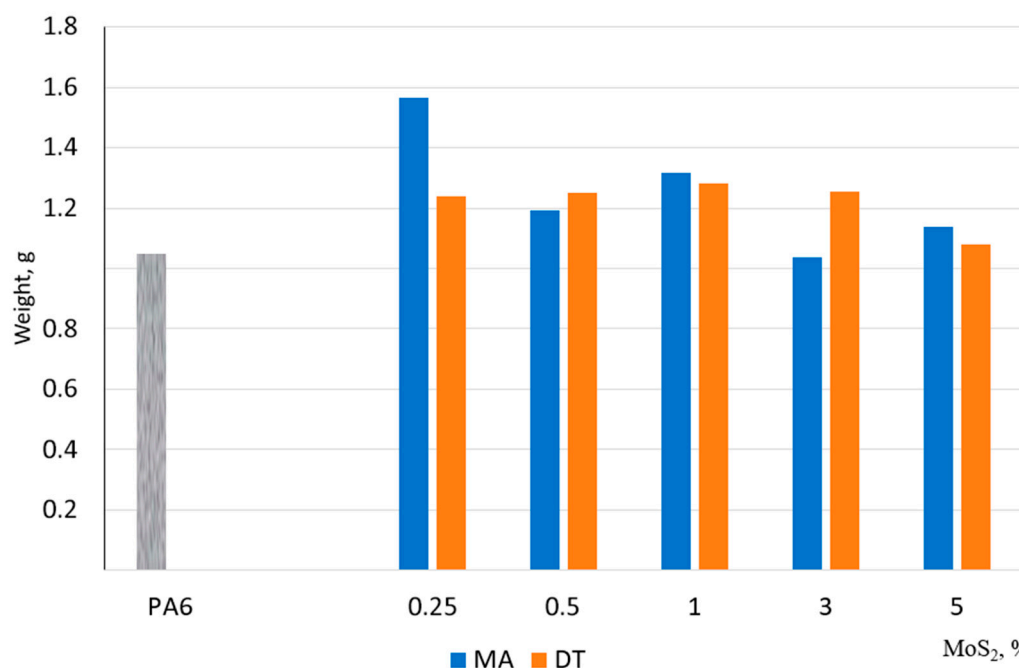


Figure 6. Dependence of water absorption of polymer composites at a temperature of 20 °C on MoS₂ content.

4. Conclusions

The paper presents a comprehensive study of the influence of mixing methods on the physicochemical, mechanical, tribological, and diffusion properties of composites based on polyamide 6 with molybdenum disulphide (MoS₂).

According to IR spectroscopy data, molybdenum disulphide interacts predominantly with oxygen-containing groups of polyamide. Mechanical activation provides stronger interaction between the matrix and filler than the classical mixing method, which is confirmed by TG/DSC analysis data. At the same time, the strength (up to 20%) and hardness (by 5%) of such composites increases. The maximum effects are observed at 3–5% filler: the modulus of elasticity increases to 1768 MPa, the relative extension decreases, and the material becomes tougher and more durable. The thermal stability of MA series composites is higher than that of composites obtained by the classical DT mixing method. The reduction in the friction coefficient demonstrates the high effectiveness of MoS₂ as a self-lubricating additive.

The introduction of MoS₂ increases the porosity and maximum water absorption of the composite by up to 20% compared to the original PA6, which is important to consider when operating in high humidity conditions.

Mechanical activation of MoS₂ enables the creation of high-strength, wear-resistant and thermally stable composites based on PA6 with a minimum content of expensive modifier (0.5–3%). The results are relevant for the production of friction elements for VR tracks, high-performance bearings and other engineering products that require a combination of low friction and high strength.

Beyond the specific case of VR platforms, the process-centred approach to interfacial engineering presented in this study is applicable to a wide range of PA6-based tribological components—bushings, guides, bearing, and sliding elements—where it is crucial to

combine low friction with the retention of strength and hardness at a minimal solid-lubricant content. The demonstrated “premixing → interphase → properties” relationship has a general character and can be extended to other lamellar or two-dimensional systems.

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Appendix A



Figure A1. Examples of damaged VR platform surfaces.

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